

Hoang Ngo

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Education

University of Florida

FL, USA

Ph.D. in Computer Science, GPA: 4.0/4.0

Expected December 2026

Dissertation Title: Learning-Based Optimization for NISQ Systems and Quantum Applications.

Vietnam National University

Vietnam

B.Sc. in Computer Science (Honors Program), GPA: 3.83/4.00 (Ranked 1st)

2020

Undergraduate Thesis: Adaptive Smart Home Lighting Control using Genetic Algorithms

Research and Work Experience

University of Florida

Graduate Research Assistant

2022–Present

- Developed reinforcement learning frameworks to address quantum hardware-aware challenges such as routing efficiency, and resource utilization.
- Conducted foundational research on privacy and security of quantum machine learning and LLMs.
- Designed and implemented quantum–classical hybrid models for real-world applications including anomaly detection, motif identification, and enzyme identification.
- Authored and co-authored peer-reviewed journal and conference publications in quantum computing and machine learning.
- Contributed to grant proposals and funded research initiatives.

VNU - University of Engineering and Technology

Vietnam

Undergraduate Research Assistant

2018–2020

- Conducted foundational research in Networking and IoT.
- Developed a smart home platform using ECHONET Lite and designed an energy-efficient smart lighting solution.

Teaching Experience

University of Florida

Graduate Teaching Assistant

- Served as a teaching assistant for *COP4533 — Algorithm Abstraction & Design* (200+ students), responsible for grading, preparing homework solutions, and assisting with exam development.

VNU – University of Engineering and Technology

Vietnam

Associate Lecturer

2020–2022

- Taught and coordinated large-enrollment courses (200+ students) including Computer Networks and Socket Programming, managing lectures, labs, and assessments.

- Co-organized the NAPROCK International Programming Contest (2020–2021).
- Mentored undergraduate students in research development and academic writing.

Publications

Journal papers.....

- **[J-5]. Hoang M. Ngo***, Nguyen H. K. Do*, Minh N. Vu, Tamer Kahveci, My T. Thai, "CHARME: A chain-based reinforcement learning approach for the minor embedding problem," ACM Transactions on Quantum Computing 7, 1, Article 2, 2026 (* indicates equal contribution).
- **[J-4]. Hoang M. Ngo**, Tamer Kahveci, My T. Thai, "FIDDLE: Fidelity Improvement through Deep Reinforcement Learning in noisy quantum systems," ACM Transactions on Quantum Computing 7, 1, Article 7, 2026.
- **[J-3]. Hoang M. Ngo**, Hong-Ha Do, Hoang-Dung Pham, Quang-Huy Do, H. S. Nguyen, "Anomaly Detection based on Frequent Pattern Mining in Smart Home Devices," IEEE Access vol. 13, 2025.
- **[J-2]. Hoang M. Ngo**, Tamim Khatib, My T. Thai, Tamer Kahveci, "QOMIC: quantum optimization for motif identification, Bioinformatics Advances, Volume 5, Issue 1, 2025, vbae208.
- **[J-1]. Hoang M. Ngo**, My T. Thai, Tamer Kahveci, QuTIE: Quantum optimization for Target Identification by Enzymes, Bioinformatics Advances, Volume 3, Issue 1, 2023, vbad112.

Conference papers.....

- **[C-5]. Hoang M. Ngo**, Nhat Hoang-Xuan, Quan Minh Nguyen, Nguyen H. K. Do, Incheol Shin, My T. Thai, "Q-ShiftDP: A Differentially Private Parameter-Shift Rule for Quantum Machine Learning," The Twenty-Ninth Annual Conference on Artificial Intelligence and Statistics (AISTATS 2026).
- **[C-4]. Hoang M. Ngo**, Ha Hong Do, Quan Dam Pham, Thinh V. Le and H. S. Nguyen, "Detecting Anomaly in Smart homes based on Mahalanobis Distance," 2024 IEEE International Conference on Communications (ICC), Colorado, USA, 2024.
- **[C-3]. Hoang M. Ngo**, Tamer Kahveci, My T. Thai, "ATOM: An Efficient Topology Adaptive Algorithm for Minor Embedding in Quantum Computing," 2023 IEEE International Conference on Communications (ICC), Rome, Italy, 2023, pp. 2692-2697.
- **[C-2]. H. S. Nguyen, X. A. Do, H. Le, V. H. Nguyen, Q. K. Duong, X. V. C. Nguyen, Hoang M. Ngo**, "ECHONET Lite-based IoT Platform for Smart Homes," 2020 RIVF International Conference on Computing and Communication Technologies (RIVF), Ho Chi Minh City, Vietnam, 2020, pp. 1-6.
- **[C-1]. Hoang M. Ngo**, X. V. C. Nguyen, Q. K. Duong, H. S. Nguyen, "Adaptive Smart Lighting Control based on Genetic Algorithm," 2019 25th Asia-Pacific Conference on Communications (APCC), Ho Chi Minh City, Vietnam, 2019, pp. 320-325.

Completed & Under Review Works.....

- **Hoang M. Ngo**, Tre' R. Jeter, Jung Taek Seo, My T. Thai, "QUPID: A Partitioned Quantum Neural Network for Anomaly Detection in Smart Grids," IEEE Transactions on Smart Grid (**Minor revision**).
- **Hoang M. Ngo**, Tre' R. Jeter, Incheol Shin, Wanli Xing, Tamer Kahveci, My T. Thai, "Guaranteeing Privacy in Hybrid Quantum Learning through Theoretical Mechanisms," The Fourteenth International Conference on Learning Representations (**Under Review ICML 2026**).
- Quan Minh Nguyen, Min-Seon Kim, **Hoang M. Ngo**, Trong Nghia Hoang, Hyuk-Yoon Kwon, My T. Thai, "Leveraging Soft Prompts for Privacy Attacks in Federated Prompt Tuning," The Fourty-Third International Conference on Machine Learning (**Under Review ICML 2026**).
- **Hoang M. Ngo**, Nhat Hoang-Xuan, Tre' R. Jeter, Thang N. Dinh, Tamer Kahveci, My T. Thai, "Quantum FACT: Full-stack Optimization for QAOA".

Honors and Awards

- University of Florida, CISE Graduate Scholarships 2023, 2024 and 2025.
- Eight scholarships for academic excellence at VNU-University of Engineering and Technology for eight semesters in a row from 2016 to 2020.
- The Valedictorian Award - Awarded for achieving the highest GPA among students graduating in the Fall semester of 2020 at VNU-University of Engineering and Technology.
- Best Bachelor Thesis Award - Awarded by VNU-University of Engineering and Technology in 2020.
- First prize in the Scientific Research Conference for Undergraduates held by the Faculty of Information Technology, VNU-University of Engineering and Technology in 2019.
- First prize in the Scientific Research Conference for Undergraduates held by VNU-University of Engineering and Technology in 2019.
- Third prize in the Scientific Research Conference for Undergraduates held by Vietnam National University, Hanoi in 2019.
- Scholarship of Tokyo Mitsubishi UFJ in 2019.
- Odon Vallet Scholarship in 2015.
- Third prize of Vietnam Olympiad in Informatics (VNOI) 2014.

Technical Skills

- **AI:** LLM Fine-Tuning, LLM Prompt Engineering, Quantum Machine Learning, Reinforcement Learning, Graph Neural Network, Gaussian Process, Federated Learning.
- **Languages:** Python, C++, Java, NodeJS, Haskell, LaTeX
- **Tools/Frameworks:** CUDA-Q, PyTorch, TensorFlow, Qiskit, PennyLane, Cirq, Tket, Gurobi, Docker, Git
- **Other:** Linux, Git, Google Cloud, Amazon Cloud

Professional Services

- Reviewed papers of the 12th International Conference on Computational Data and Social Networks (CSoNet 2023).
- Participated in hosting the NAPROCK International Programming Contests in 2020 and 2021.

References

- **Dr. My T. Thai**
Professor, Department of Computer & Information Science & Engineering
University of Florida
Email: mythai@cise.ufl.edu
- **Dr. Tamer Kahveci**
Professor, Department of Computer & Information Science & Engineering
University of Florida
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